

# MAC 233: Servo Lab

Dr Max Champneys

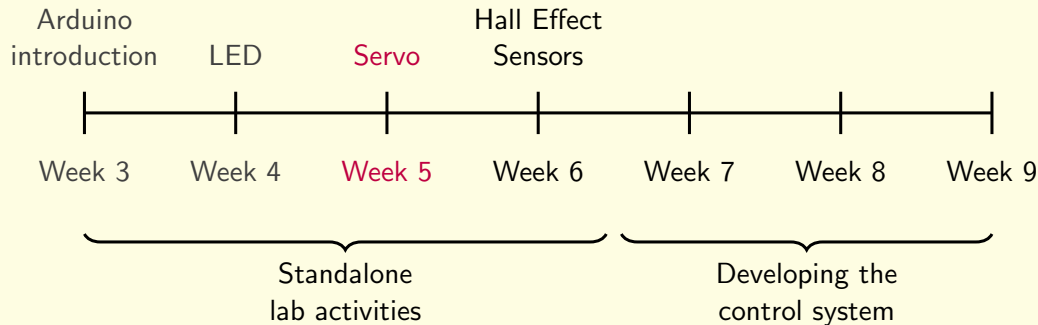
School of Mechanical, Aerospace, and Civil Engineering,  
University of Sheffield, UK

MAC 233: **28/10/2026**



University of  
**Sheffield**

# MAC 233: LABS OVERVIEW



## TODAY'S LAB: PREREQUISITES

This worksheet builds upon the theory of how to use an external pin that was covered in the Lab 2 worksheet. Whilst you won't need any code from Lab 2, you will need to have covered how to use a breadboard and how to wire up an external output to an Arduino pin.

Out of your Arduino pack, you will need: 1 Arduino, 1 breadboard, 1 servo, and 3 jumper cables.

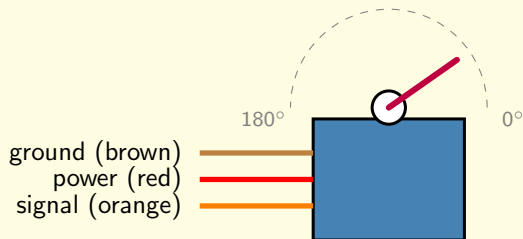
# TODAY'S LAB: OBJECTIVES

The objective of today's lab is to learn how to control a servo using the inbuilt Arduino Servo Library:

<https://docs.arduino.cc/libraries/servo/>.

The worksheet will take you through the steps of how to connect an external servo to your Arduino and create a sketch that rotates the servo in increments of 30 degrees.

# WHAT IS A SERVO?



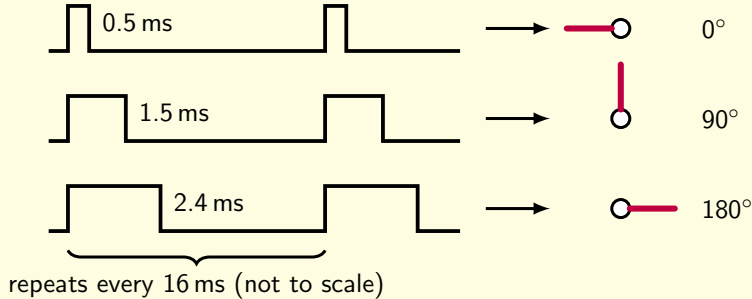
A servo is a motor with a gearbox and a small control circuit built into it. You tell it an *angle*, and it turns to that angle and holds it there.

Two of its wires just supply power. The third carries the signal that tells the servo which angle you want.

# NEW: THE SERVO LIBRARY

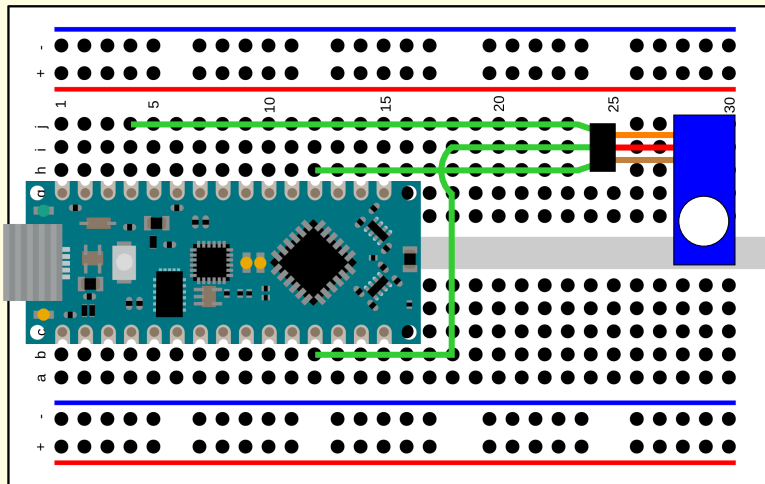
- ▶ `#include <Servo.h>`  
makes the library available to your sketch
- ▶ `Servo servo;`  
creates a servo for you to control
- ▶ `servo.attach(SERVO_PIN);`  
says which pin the signal wire is on — once, in `setup`
- ▶ `servo.write(90);`  
moves the servo to an angle between 0 and 180 degrees

# WHAT 'SERVO.WRITE' ACTUALLY DOES



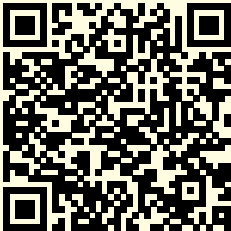
The Servo library sends a short 5 V pulse every 16 ms, and the *width* of that pulse tells the servo which angle to turn to.

## WIRING DIAGRAM



Breadboard source: <https://freesvg.org/breadboard>.

Lab 3 worksheet



Nano Every pinout



Problems? Raise your hand and someone will come and help you.